Classification of At-Risk Counties during COVID-19

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➢ Executive Summary

This project addresses the critical need to classify counties in Texas based on their vulnerability to COVID-19, providing a data-driven framework to guide public health interventions. Understanding which counties are at higher risk is essential for allocating resources, implementing targeted policies, and mitigating the impact of future outbreaks. The study integrates diverse datasets—including census demographics, vaccination rates, and mobility patterns—to calculate a comprehensive risk score for each county. The model identifies high-risk counties, which often exhibit factors like high population density, low vaccination rates, and significant past COVID-19 impacts. Three classification models—k-NN, Naive Bayes, and Neural Network—were evaluated. Visualizations, such as risk maps and ROC curves, confirm the model's effectiveness in distinguishing risk levels, particularly for low- and high-risk counties (AUCs of 0.979 and 0.907, respectively). The results are invaluable for policymakers and public health officials, enabling proactive resource allocation, targeted vaccination campaigns, and tailored public health measures. By identifying at-risk counties, the model supports data-driven decision-making to reduce the burden of future pandemics.

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1 Data Preparation

The data used in this project is much greater than previous projects, where looking at 5–10 different variables would offer a good insight into visualizations and clustering analysis. Here, I chose to use many variables from Census, Mobility, and Vaccine data to describe trends in counties to properly classify them. To see how well a county is prepared in the event of another pandemic, it is not as simple as looking at responsiveness to mobility and vaccine mandates, but risk can be derived from how the community itself handled the entirety of COVID-19. First, before explaining visuals or trends, the data must be cleaned and processed into a form capable of being classified.

For data quality and missing values, a few steps were taken to allow classifiers that require no missing values to function properly. For most features, missing values were omitted during their creation so that there would be no issue with merging down the line. Another problem found was that some rows in the data had no county or cases associated with them, which were promptly removed as they provided no insight. Another action against missing values was that counties had many dates in which data was gathered, so instead of removing an entire county from the database, one of the many dates was removed, preserving the county's overall data during the entire pandemic. This data was later merged into one entry per county by taking the average and normalizing. Not only did this method protect the data quality against missing values, but also against outliers, since they are all compared with counties with similar attributes.

1.1 Census Data

Feature	Scale of Measurement	Description
Total_pop	Ratio	Total population of individuals living in the county.
Median_income	Ratio	Median income of individuals living in the county.
Median_age	Ratio	Median age of individuals living in the county.
Median_year_structure_built	Interval	The median year when residential structures in the area were built.
Households	Ratio	Number of households within the county.
Nonfamily_households	Ratio	Number of households not composed of people related by birth, marriage, or adoption.
Housing_units	Ratio	Total number of housing units in the area.
Vacant_housing_units	Ratio	Number of housing units that are unoccupied.
Percent_income_spent_on_rent	Ratio	The average percentage of household income spent on rent.
Employed_pop	Ratio	Number of people are currently employed.
Unemployed_pop	Ratio	Number of people are currently unemployed.

Table 1: Census Data metrics used for classification

For the population characteristics, I used the total population of the county to normalize all other variables that had to do with the relation to population. For instance, population density and per capita rates for cases and deaths were calculated by the total population. This is so that specific counties will not be a heavy outlier to the rest of the data, and that all the county's characteristics are relative to its population. The median age was kept as the original values since it was already a median. Higher values of age contributed positively to demographic vulnerability.

It is important to investigate the economic factors because they can shed insight into the current state and quality of life in the county. Median income was unchanged but was averaged when aggregating at the county level and affects the risk level by an inversely Z-score standardization. I chose to invert the Z-score because generally higher income allows citizens to pay for better prevention and healthcare to combat the virus. The percentage income on rent variable was used to investigate demographic vulnerability considerations since a higher rent generally means that there is a higher quality of living. Lastly, the employed and unemployed population was considered. Each was normalized to calculate the unemployment rate, which would lead to a lower income, increasing demographic vulnerability.

Housing and living conditions were used to research the transmission risk for individuals and families. Things like housing density were taken into consideration due to the closer people are together the higher the risk of transmitting the virus. Vacant and nonfamily housing also contribute to the density of the housing situation in each county. Lastly, the median year built gave a better insight into the infrastructure quality, allowing the viewer to understand what kind of residential, commercial, and government buildings are present in the county, as older buildings can have poorer building codes and are possibly outdated.

1.2 Vaccine Data

Avg_First_Dose_Pct	Ratio	People with at least one Dose by State of Residence
Avg_Full_Dose_65Plus	Ratio	Percent of people ages 65+ who have completed a primary series (have second dose of a two-dose vaccine or one dose of a single-dose vaccine) based on the jurisdiction where vaccine recipient lives
Avg_Full_Dose	Ratio	Percent of people who have completed a primary series (have second dose of a two-dose vaccine or one dose of a single-dose vaccine) based on the jurisdiction and county where vaccine recipient lives

Table 2: Vaccination Data metrics used for classification

The vaccine data was prepared so that the average dosage percentage was clear for each county. This data was considered on how to calculate the total risk. As learned from previous projects, the vaccines did work and provided a higher chance of survival and helped lower hospitalization rates. So, the higher the vaccination rate of the county, the less at risk it is.

1.3 Mobility Data

Mean_work_Pct	Interval	The overall change in workplace mobility during COVID-19
Sd_work_pct	Ratio	Volatility or variability in workplace mobility.
Min_work_pct	Interval	The most extreme drop in workplace visits during COVID-19 for the county.
Max_work_pct	Interval	The most extreme increase in workplace visits during COVID-19 for the county.
Mean_retail_Pct	Interval	Shows the average change in traffic to recreational venues compared to baseline.
Sd_retail _pct	Ratio	Reflects the volatility in visitation to these venues.
Min_retail _pct	Interval	Captures the lowest point of visitation
Max_retail _pct	Interval	Captures the highest point of visitation
In_school	Ratio	The number of students that are currently in school.
In_undergrad_college	Ratio	Number of students that are currently in undergrad college.
Commuters_by_public_transportation	Ratio	Number of residents who commute to work using public transit.

Table 3: Mobility Data metrics used for classification

Mobility data dealing with the change in the workplace and retail mobility was explored further. Because mobility fluctuates during the pandemic because of different mutations of the virus or mandates from the government, these variables include the mean, standard deviation, minimum, and maximum to better grasp the whole story. Commuters and schooling were normalized by dividing it by the population of the county and used to calculate the transmission risk. Schools and public transportation are huge hubs of disease spread, so each had to be carefully incorporated into the risk model.

1.4 Data Derived for Risk Calculation

Death_per_case	Ratio	The number of deaths per case in the county
Deaths_per_10000	Ratio	The number of deaths due to COVID-19 for every 10000 people in the county
Cases_per_10000	Ratio	The number of cases of COVID-19 for every 10000 people in the county
Risk_Level	Ordinal	The prediction of how dangerous another pandemic will be to the county
Historical_Impact	Interval	The severity of COVID-19's past impact on a county

Demographic_Vulnerability	Interval	How vulnerable a county's population is to severe COVID-19 outcomes based on demographic factors
Transmission_Risk	Interval	How conducive a county's conditions are to virus transmission
Total_Risk_Score	Interval	The overall pandemic vulnerability assessment

Table 4: Data used as extra insight to calculate county risk

In this table are the very important resulting variables calculated along the way. These provide the greatest insight from the other data, showing how deadly the virus was in each county, and now quantifying the impact of COVID-19. The demographic vulnerability includes factors like age or income. Transmission risk involves factors describing how much contact a person has with another person. The historical impact is a general idea of how the county dealt with COVID-19 across the entire pandemic, pulling from the cases and deaths. In the next section, the process of how low, medium, and high-risk counties are calculated will be explained.

2 Calculating Risk

Initially, when compiling the data to classify, risk was only defined as how many lethal the virus was in the county without considering the factors that led to the conclusion. However, this would have been a poor classification of risk, and when tested, had very little data to offer. Classifiers did not analyze the different pieces of data that were included and jumped to death tolls. With this new classification of risk, each classifier used is much more accurate and paints a much larger and in-depth picture.

As mentioned previously, the risk calculation uses this raw data: The cases and death per 10000 population, death per case ratios, normalized demographic features relative to population, vaccination coverage percentages, and Mobility pattern summaries. Three major risk components were created from the z-scores.

1. Historical Impact (40% weight): death rates, case rates, fatality rates
2. Demographic Vulnerability (30% weight): median age, unemployment, income (inverse)
3. Transmission Risk (30% weight): population density, public transit use, housing density, vaccination rates (inverse)

Historical Impact has a higher weight than the rest because past patterns of disease spread and mortality are strong predictors of future vulnerability. Areas that experienced high case or death rates previously have demonstrated vulnerability that may persist through new waves of infection. High historical impact results from a combination of demographic, healthcare, socioeconomic, and behavioral factors working together, and many of these are not directly measurable in the data. Historical evidence is a proven indicator of how severely COVID-19 has already affected a county, making them highly reliable predictors of vulnerability.

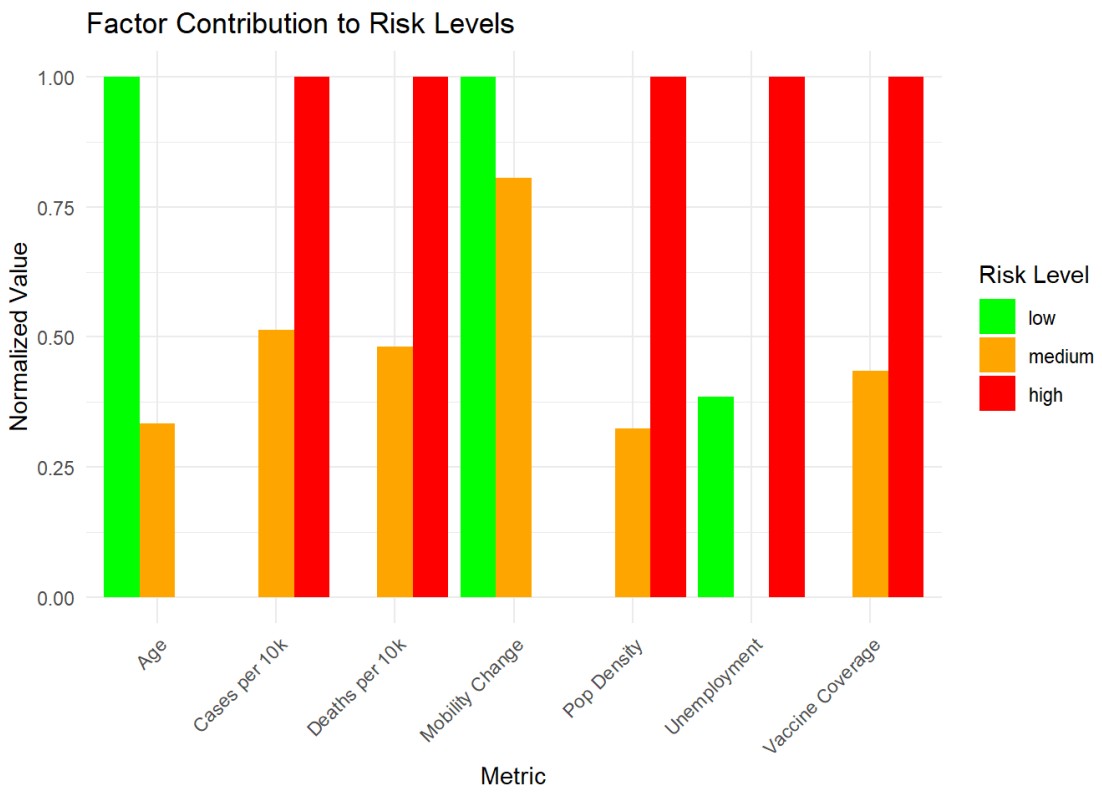

Table 5: Major Factors Contributing to Risk Level

These are the major contributors to calculating risk levels. High risk factors include cases, deaths, population density, unemployment, and vaccine coverage. Cases, deaths per 10k, and population density is obvious why they contribute to high risk, but interestingly, unemployment and vaccine coverage also contribute much to defining high risk. Vaccines are made to combat the virus and have worked to dampen the effect of the virus, so why are heavily vaccinated counties higher risk than others?

Vaccines were often prioritized in already high-risk areas. Counties with severe outbreaks likely received vaccines earlier and in greater quantities, creating a correlation between high vaccination rates and high pre-existing risk. Many private businesses and schools required the vaccine to physically come into work or class, so people were coerced into getting the virus regardless of whether the county was considered “high risk” or not. Another possibility to this paradoxical result is that rather than vaccination causing higher risk, the causality may run in the opposite direction - higher risk areas pursued more aggressive vaccination campaigns. This finding doesn't mean vaccines are ineffective - at the individual level, vaccination dramatically reduces risk of severe illness and death. Rather, it highlights the complex relationship between population-level vaccination rates and overall community risk profiles that incorporate multiple historical and demographic factors.

On the other side of the spectrum, low risk counties may be older and more compliant with mobility restrictions. Older populations generally have lower mobility rates and fewer daily contacts than younger populations, especially in educational and workplace settings where superspreader events often occur. Older populations

demonstrated greater risk awareness during the pandemic and higher compliance with protective measures like masking, distancing, and isolation recommendations. This could be why Unemployment can contribute to both low and high risk, if people are unemployed then they might be less mobile, while on the other hand they might be more mobile, unemployment can contribute to both, which is why there are many metrics to determine where the risk is in the state. Lastly, historically, counties with larger older populations most likely received prioritized healthcare resources, testing, and earlier vaccination campaigns

3 Modeling

3.1 Modeling Risk

Now that risk has properly been defined and explored, models can be constructed under good faith that they reflect the qualities inherent with low, medium, and high-risk counties.

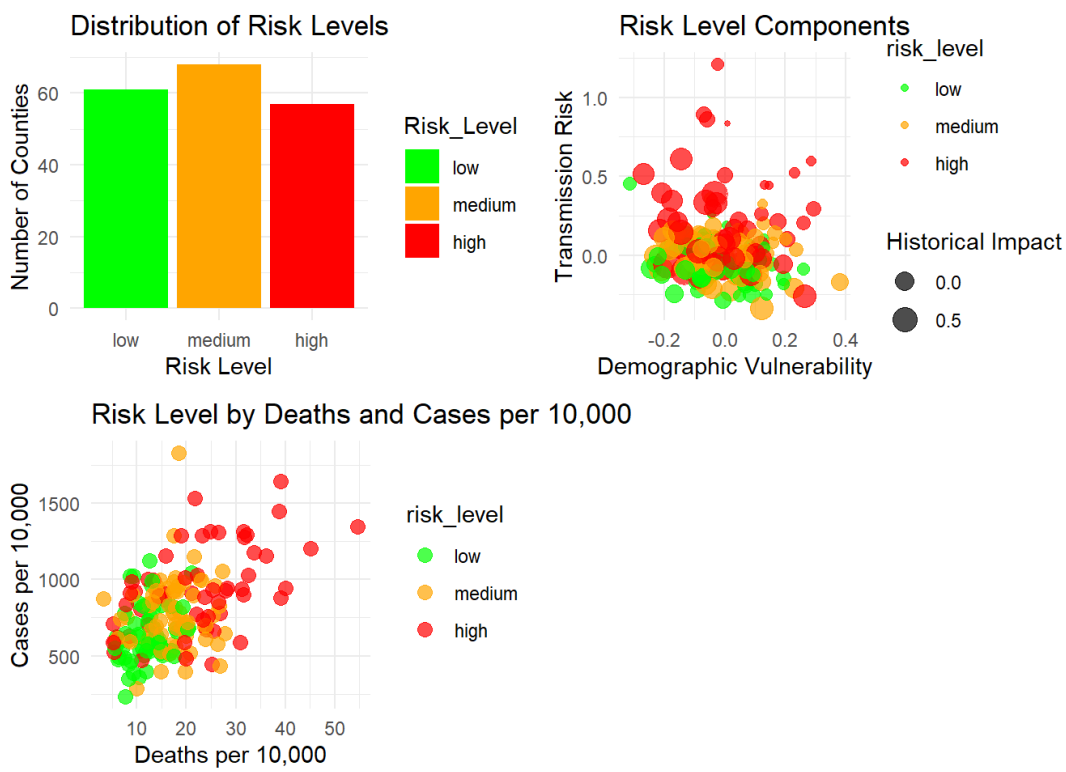

Table 6: County Distribution by Risk

The first plot showing the Distribution of Risk Levels shows that the levels are balanced where most of the counties are considered a medium risk level, and the least are high risk. This outcome is good so that not one category contains much more than the others. If they were more unbalanced, the risk calculations would probably need to be reconsidered.

The scatterplot with Risk Level Components explains a lot of what is going on behind the scenes. To begin, there are the three risk levels, plotted on a graph comparing transmission risk and demographic vulnerability. The relation is not linear, which is a safe indication that each county's features are being considered, not just one factor. The historical impact is also present, and counties that were hit the hardest can be identified by larger points on the graph. Most of the larger points sit outside of the bubble where the three mix. This interesting fact explains how the historical impact is a higher indication of how at risk a county is.

Lastly, the final scatterplot shows the Risk Level by Deaths and Cases per 10k. Counties that had higher cases and deaths as a result are classified as higher risk. Some high-risk counties had less cases and deaths than some low-risk counties, which may seem counterintuitive, but after accounting for smaller counties where although it is less populated, COVID-19 spread just as much if not faster in these communities.

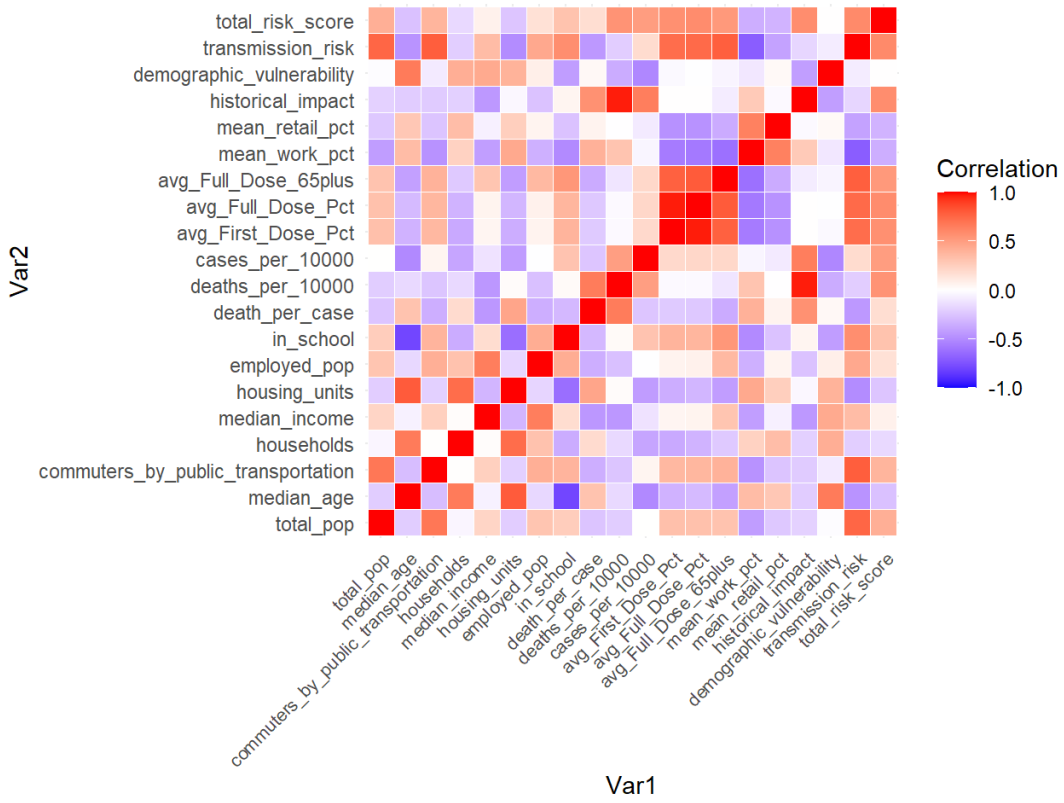

Table 7: Correlation of Risk Factors

Here, we have the risk score components. The high correlation between total risk score and its components validates the composite risk score construction in the model. As for the COVID-19 Outcome variables, there is another high correlation between deaths and cases per 10,000 and historical impact. Deaths per case shows correlation with historical impact but less with raw case counts. For the vaccination impacts, there is a negative correlation between vaccination metrics and transmission risk, which suggests counties with higher vaccination rates tend to have lower transmission risk. Leading into mobility patterns, the mean retail and work percent change show a correlation with each other. They have negative correlation with some risk factors, this suggests that counties with less mobility

reduction may have a higher risk of COVID-19. For the demographic factors, we can see that there is a negative correlation between median income and demographic vulnerability. Higher income can mean that there is a chance of higher quality healthcare if there are to be an outbreak. Another metric here is the correlation between commuters by public transport and transmission risk. The more people interact on public transport naturally means there is a higher risk for transmission. For population specific variables, the counties with higher median age show some correlation patterns with vulnerability metrics. Lastly, there is a high correlation between employed population and the in-school population.

The heatmap provides evidence for the underlying assumptions in the risk model - higher population density, lower vaccination rates, lower income, and higher use of public transportation all contribute to increased COVID-19 risk factors. The correlations support the multi-dimensional approach to risk assessment using historical impact, demographic vulnerability, and transmission risk components. This correlation analysis helps validate the risk classification model and identifies which factors most strongly influence county risk levels in Texas.

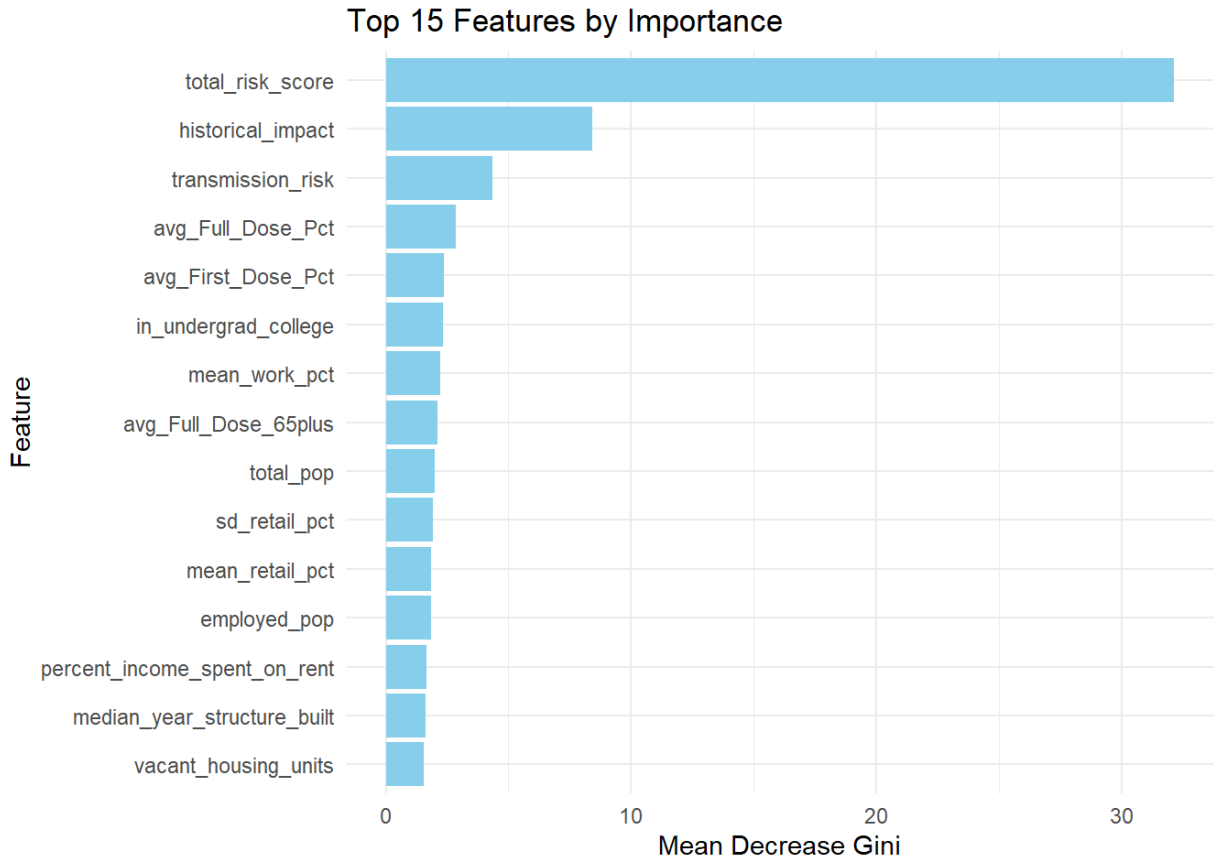

Table 8: Classification Model Gini Index

The Mean Decrease Gini shown in the plot is a measure of feature importance used in decision tree-based models. It quantifies how much each feature contributes to reducing impurity in the model—essentially, how effective each feature is at helping the model make better decisions when splitting data. Features with a higher Mean Decrease Gini are more important because they result in greater reductions in classification error across the trees in the forest.

In the plot, the feature total_risk_score has the highest importance by a significant margin, indicating that it plays the largest role in driving accurate predictions. This means the model relies heavily on this feature to distinguish between different classes or outcomes. Other features like historical_impact and transmission_risk also contribute meaningfully, but to a lesser extent. Meanwhile, features with much lower Mean Decrease Gini values, such as vacant_housing_units or median_year_structure_built, have relatively little influence on the model's performance. This is intentional, since the goal was to have the total_risk_score be the whole story, but there is still some effect of the smaller "less important" variables.

3.2 Model Comparison

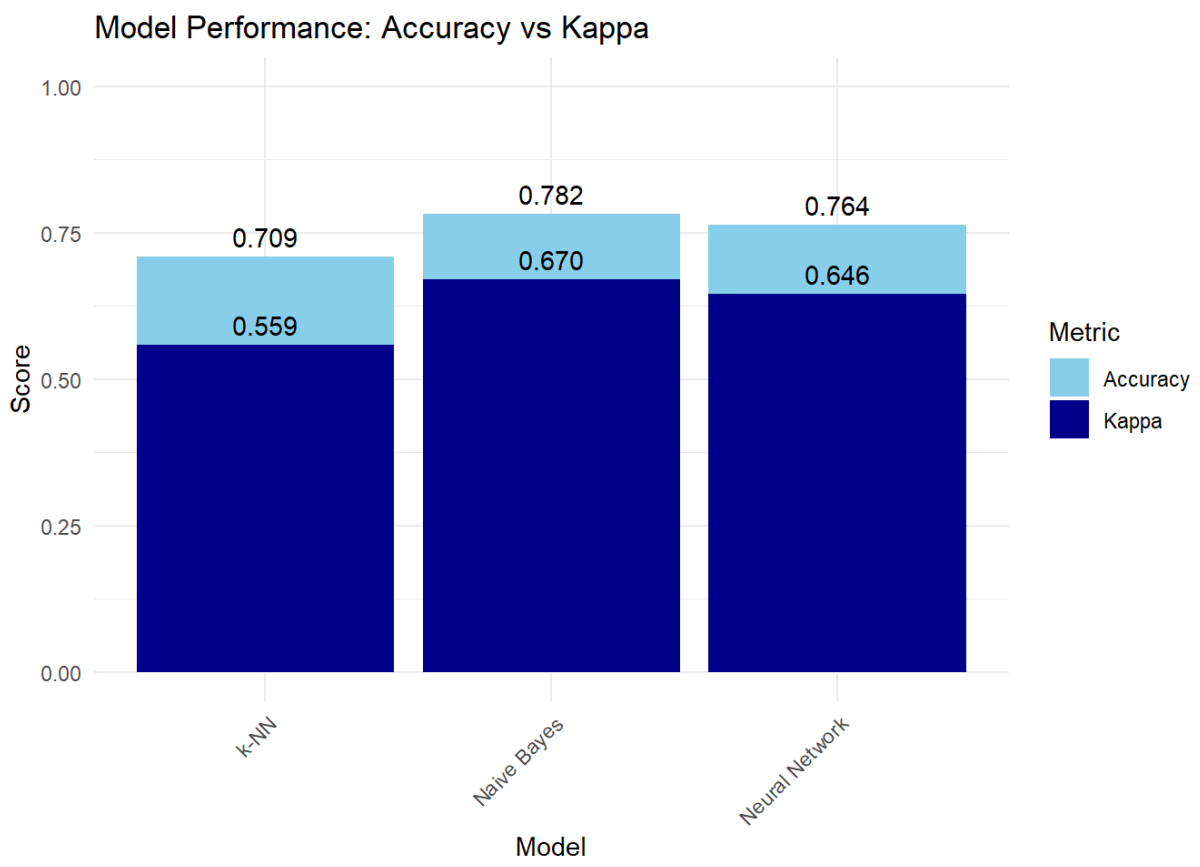

Table 9: Model Performance According to Kappa

This bar chart compares the performance of three different classification models: k-NN, Naive Bayes, and Neural Network by using two evaluation metrics: Accuracy and Kappa. Accuracy measures the proportion of correct predictions out of all predictions. Kappa accounts for the possibility of an agreement occurring by chance. A higher Kappa value indicates better model performance beyond random guessing. From the chart, Naive Bayes achieved the

highest accuracy at 0.782, followed by Neural Network at 0.764, and then k-NN at 0.709. The same ranking holds for the Kappa values: Naive Bayes leads with 0.670, Neural Network follows at 0.646, and k-NN trails at 0.559. This means that Naive Bayes not only predicted the most outcomes correctly but also did so with the most meaningful agreement beyond random chance. In summary, Naive Bayes outperformed the other two models on both metrics, making it the most effective of the three in this classification task. The inclusion of both metrics gives a more complete view of model performance, especially in cases where class distributions might be imbalanced.

The Naive Bayes classifier performs the best overall, with an accuracy of 78.2% and a Kappa of 0.670. This means that it correctly predicts the risk level of a county nearly 8 out of 10 times and does so with a strong level of agreement beyond what would be expected by chance. This means that it's not just guessing well, rather it's learning meaningful patterns from the data that correspond to real differences in county-level risk. For Naïve Bayes as a whole, it has many strengths when compared to other models, such as its simplicity and speed. Naïve Bayes can be trained and predict data very quickly and can perform very well on high and low-dimensional data. Not only can it make predictions using varying sizes of data, but it is also robust to irrelevant features, as they average out and do not drastically affect performance. Referring to Table 8, it uses the most relevant features. With these strengths also come weaknesses, as it assumes strong independence. Naïve Bayes assumes that features are conditionally independent given the class, which is rarely true in data. It also can underperform when features are highly correlated or interact in complex ways, which makes it less flexible than models like neural networks.

The Neural Network also performs well, with slightly lower accuracy (76.4%) and Kappa (0.646), indicating it's a solid alternative but slightly less reliable than Naive Bayes for the dataset. What sets the Neural Network apart from other models is that it uses neurons that reflect the human brain, and pieces together patterns in the data. It has high predictive accuracy, capable of modeling complex non-linear relationships and interactions between variables. It can also learn abstract and high-level features directly from raw data, reducing the need for manual feature engineering. However, the cost of the great strengths of the neural network is that it has a "black box" nature, making it hard to interpret or explain how the model is arriving at decisions. Not only that, but it requires a huge amount of data and is computationally expensive, luckily this is not a problem since the mobility, vaccination, and census data sets are very large in their own respect.

The k-Nearest Neighbors (k-NN) model performs the worst of the three classifiers in the evaluation. With an accuracy of 70.9% and a Kappa of 0.559, it is still better than random guessing, but noticeably less effective than both Naive Bayes and the Neural Network. This performance suggests that k-NN struggles more to distinguish between different risk levels across Texas counties. There are a few likely reasons for this. First, k-NN considers all features equally when calculating distance. Since the dataset has features that don't contribute much to distinguishing risk levels, (as seen in Figure 8), they can dilute the signal and hurt performance. Unlike Naive Bayes or Neural Networks, k-NN doesn't "learn" patterns, it simply compares test samples to training examples. That makes it vulnerable to local anomalies and less robust in complex or high-dimensional data spaces. It does not require a training phase, does not

make assumptions, is very effective on well-separated data, and works well with the multi-class model that has been chosen. Some weakness of the k-NN algorithm is that it is computationally expensive during prediction time. It can also struggle with high-dimensional data and is more sensitive to noisy and irrelevant features.

Taken together, these results suggest that the classifiers—especially Naive Bayes—are doing a good job at distinguishing high-risk vs. low-risk counties in Texas. The strong Kappa values also imply that the models are not just overfitting to dominant classes, but are finding real, generalizable patterns. This adds confidence to using the model's predictions to guide resource allocation, intervention, or further analysis.

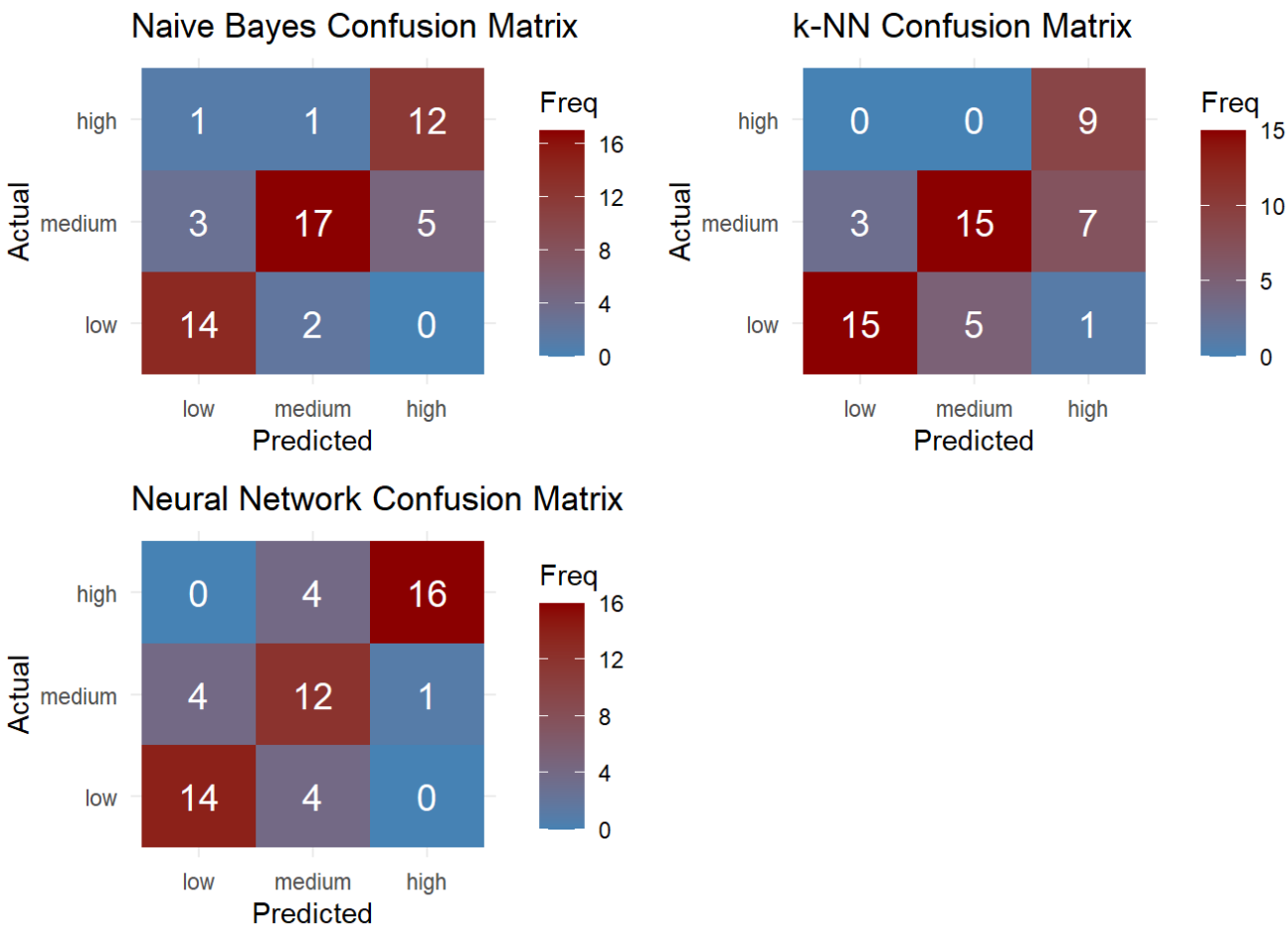

Table 10: Actual vs Predicted Classes

These three confusion matrices show the accuracy of the classification compared to the true risk label. A perfect classification of the data would have actual and predicted scores line up along the diagonal, indicating that all counties were classified into their correct risk categories.

For the Naive Bayes classifier, most predictions were accurate, particularly for low and medium-risk counties. However, there was some confusion between medium and high risk categories — five medium-risk counties were misclassified as high-risk, and one high-risk county was labeled as medium. This suggests the model may slightly

overestimate the risk, which is generally safer in public health contexts, but could lead to unnecessary resource allocation.

For the k-Nearest Neighbors classifier, a key strength is that no high-risk counties were misclassified. This is critical in a public health context, as high-risk areas require the most attention and intervention. Misclassifying them could result in delayed or absent mitigation efforts, putting populations at risk. However, k-NN struggled with low and medium-risk counties, misclassifying 5 low-risk as medium, and 7 medium-risk as high. This indicates less overall reliability, though its conservative bias in not underestimating risk is noteworthy.

The Neural Network showed strong performance overall, particularly in correctly identifying high-risk counties, with 16 out of 20 correctly classified and none misclassified as low risk. This reinforces its suitability for high-stakes health decisions. Additionally, its performance on medium and low-risk categories was solid, though a few medium and low-risk counties were misclassified into each other. This model appears to strike a balance between caution and accuracy.

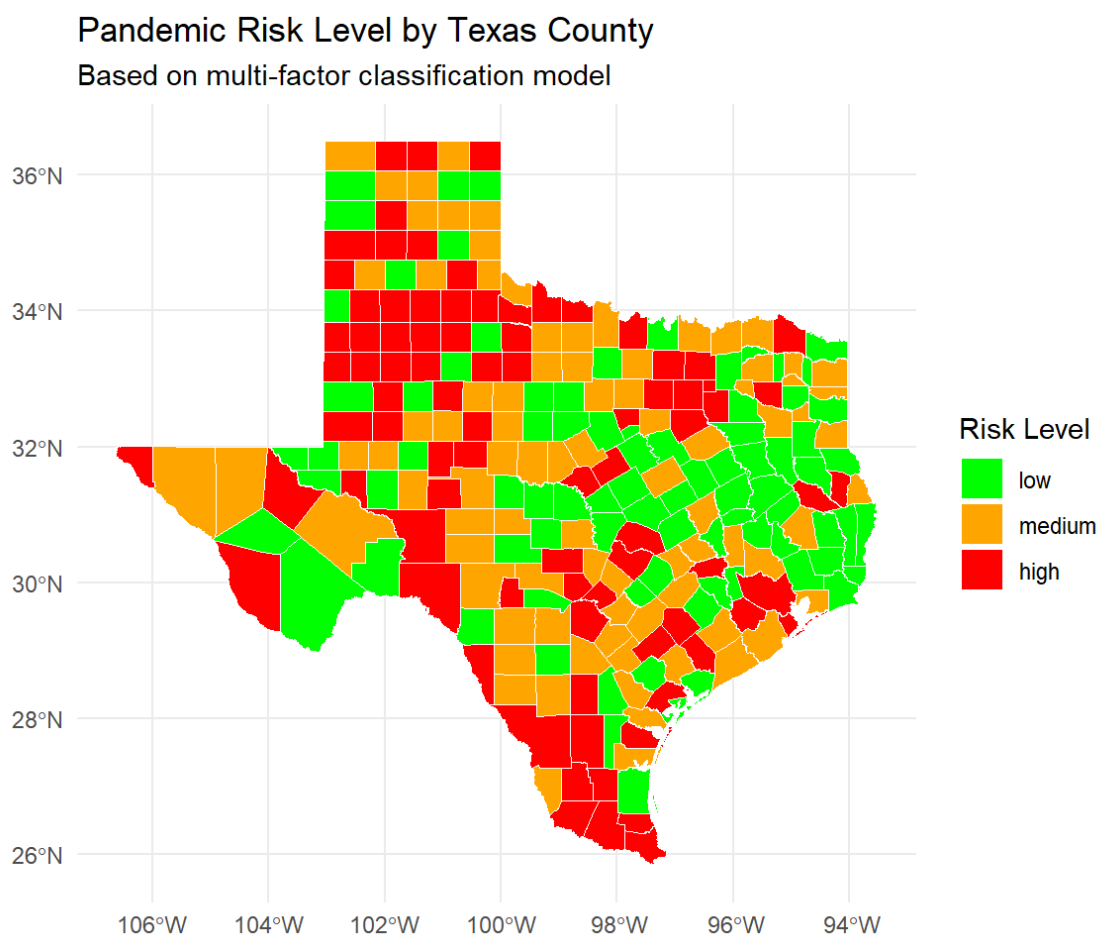

Table 11: Texas Map Divided into Separate Risk Levels

The map presents the pandemic risk classification for Texas counties based on a multi-factor model, using color to indicate the level of risk: green for low, orange for medium, and red for high. From the visual distribution, it's evident that high-risk counties are predominantly located in the Panhandle, parts of West Texas, the southern border, and certain areas in Central Texas. In contrast, lower-risk counties are more common in East and Southeast Texas, as well as near some major urban centers. This pattern suggests that the model may be capturing geographic disparities linked to healthcare access, population density, infrastructure, or socioeconomic factors. The spatial coherence of risk levels indicates that neighboring counties often share similar classifications, which may reflect regional trends in pandemic-related vulnerabilities. Overall, the visualization highlights how risk varies across the state and invites further investigation into the specific variables influencing these classifications.

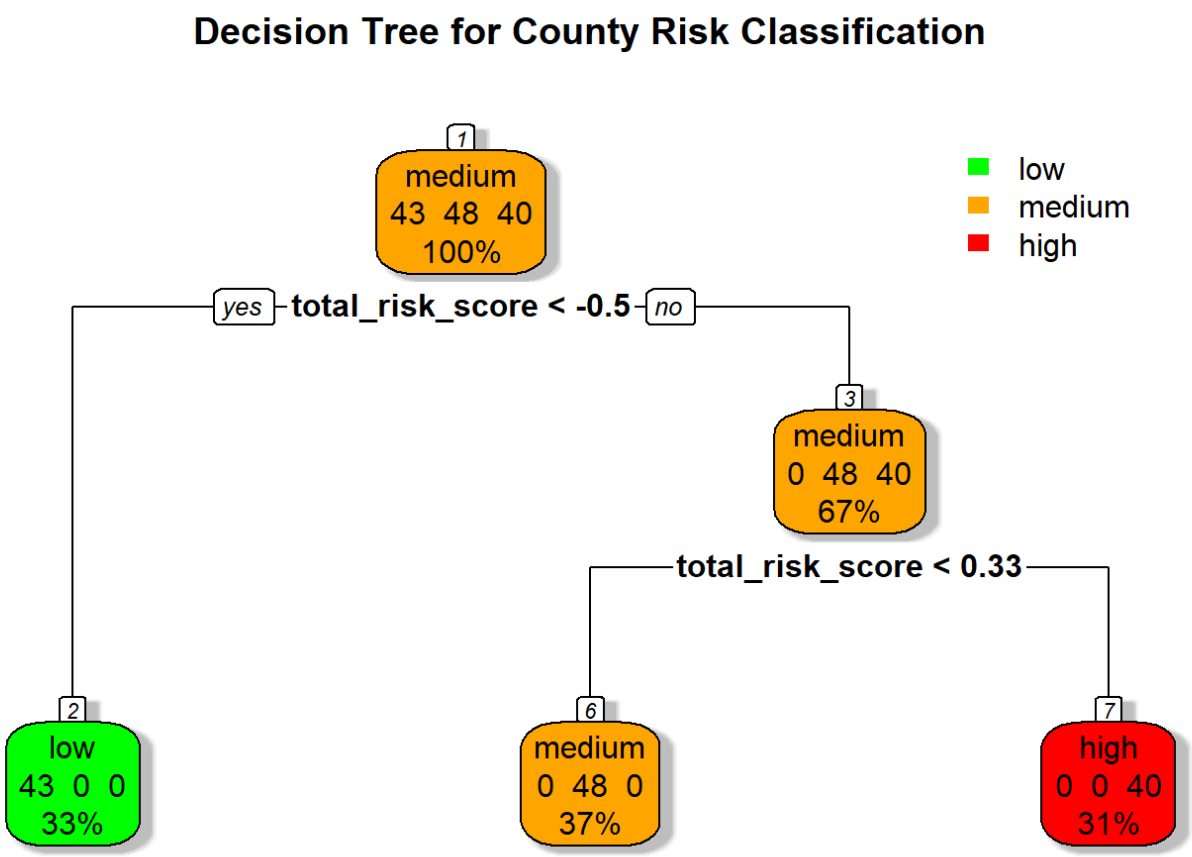

Table 12: Decision Tree for County Risk Classification

This visual shows the decision tree used by the classifiers to determine the risk of each Texas county. Each node represents a decision based on the total risk score. Root node 1 shows all counties beginning with a default medium classification, and creates a distribution of 43 low risk, 48 medium risk, and 40 high risk, totaling to 131 different counties. At the first split where the total_risk_score < -0.5, the counties with score less than that are classified as “low risk.” In the next split where the total risk score is less than 0.33, it creates the medium

classification, and the rest of which are assigned “high risk.” The terminal nodes consist of 33% low, 37% medium, and 31% high. The decision tree provides a simplified but interpretable view of how the complex classification system works.

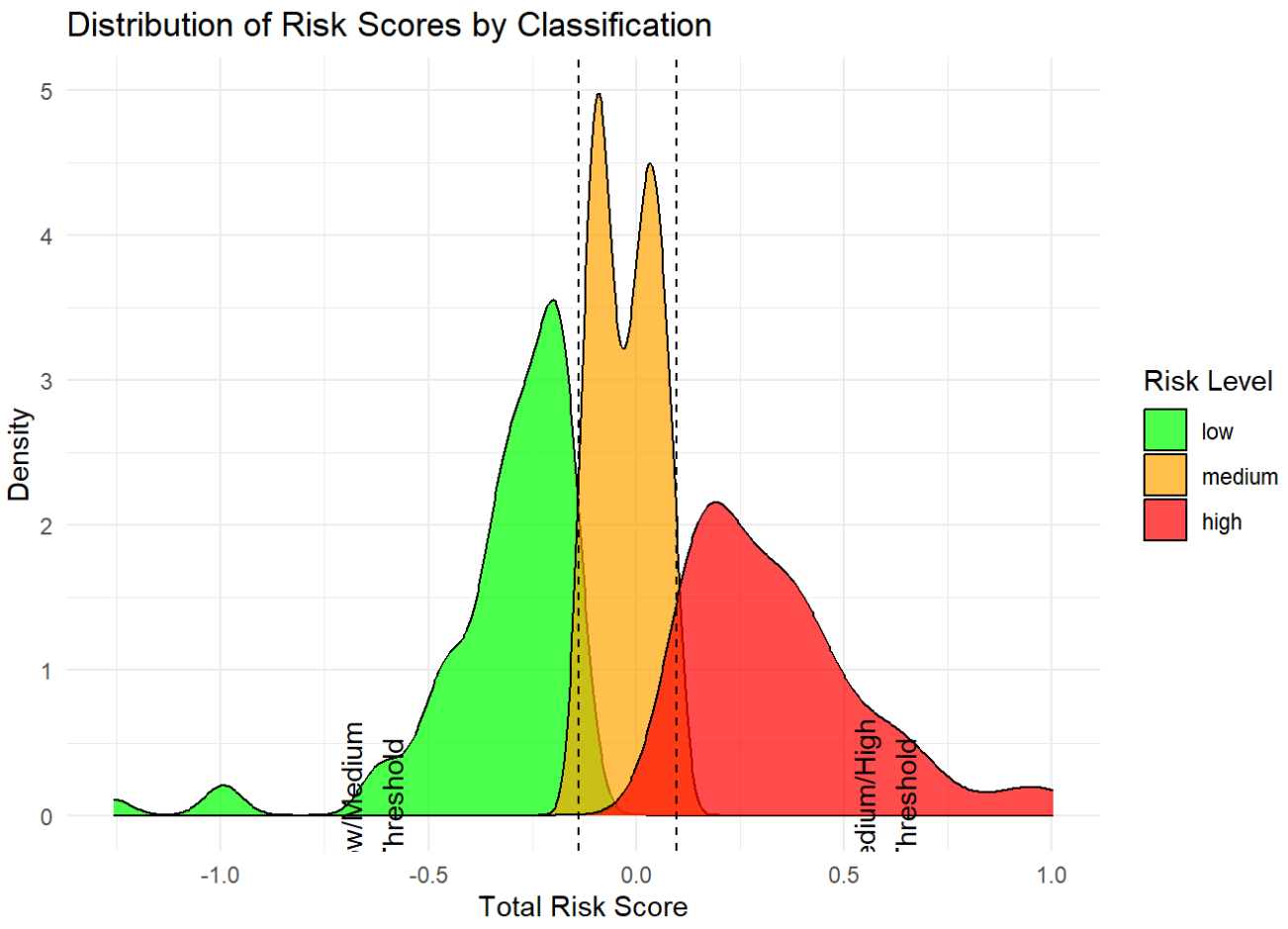

Table 13: Distribution of Risk Scores by Classification

The graph illustrates how Texas counties have been classified into COVID-19 risk categories based on a continuous scoring system. The scores range approximately from -1.0 to 1.0, with vertically dashed lines marking the thresholds that separate the risk levels. Counties categorized as low risk (green) tend to have risk scores concentrated below 0.0, peaking around -0.25, indicating relatively favorable conditions across the measured risk components. Medium-risk counties (orange) are centered around 0.0 and display a bimodal distribution, suggesting a more diverse group with overlapping characteristics from both lower and higher risk areas. High-risk counties (red) are largely found above the 0.0 threshold, with the greatest density between 0.25 and 0.75, reflecting elevated vulnerabilities and past COVID-19 impacts. This risk score distribution confirms that the classification system effectively differentiates counties based on their overall risk, while still capturing the nuance and variability at the boundaries. In the context of COVID-19, this approach enables a data-driven, tiered understanding of which areas may require more urgent intervention, resources, or continued monitoring, helping public health efforts stay responsive and targeted.

The overlap of risk scores between categories, especially near the threshold boundaries, means that some counties have similar overall risk levels despite being placed in different classification groups. In practical terms, this suggests that a county near the cutoff between, say, medium and high risk might share characteristics with counties in both categories. This overlap occurs because risk is assessed on a continuous scale, while classifications are discrete, forcing counties into distinct groups even when their scores are very close.

In the context of COVID-19, this overlap highlights the gray areas in risk assessment, where small differences in data, such as a slight increase in transmission rates or past death counts, could shift a county from one category to another. It underscores the importance of not treating risk categories as absolute but instead understanding them as part of a broader continuum, where counties near the edges may warrant flexible or closely monitored responses.

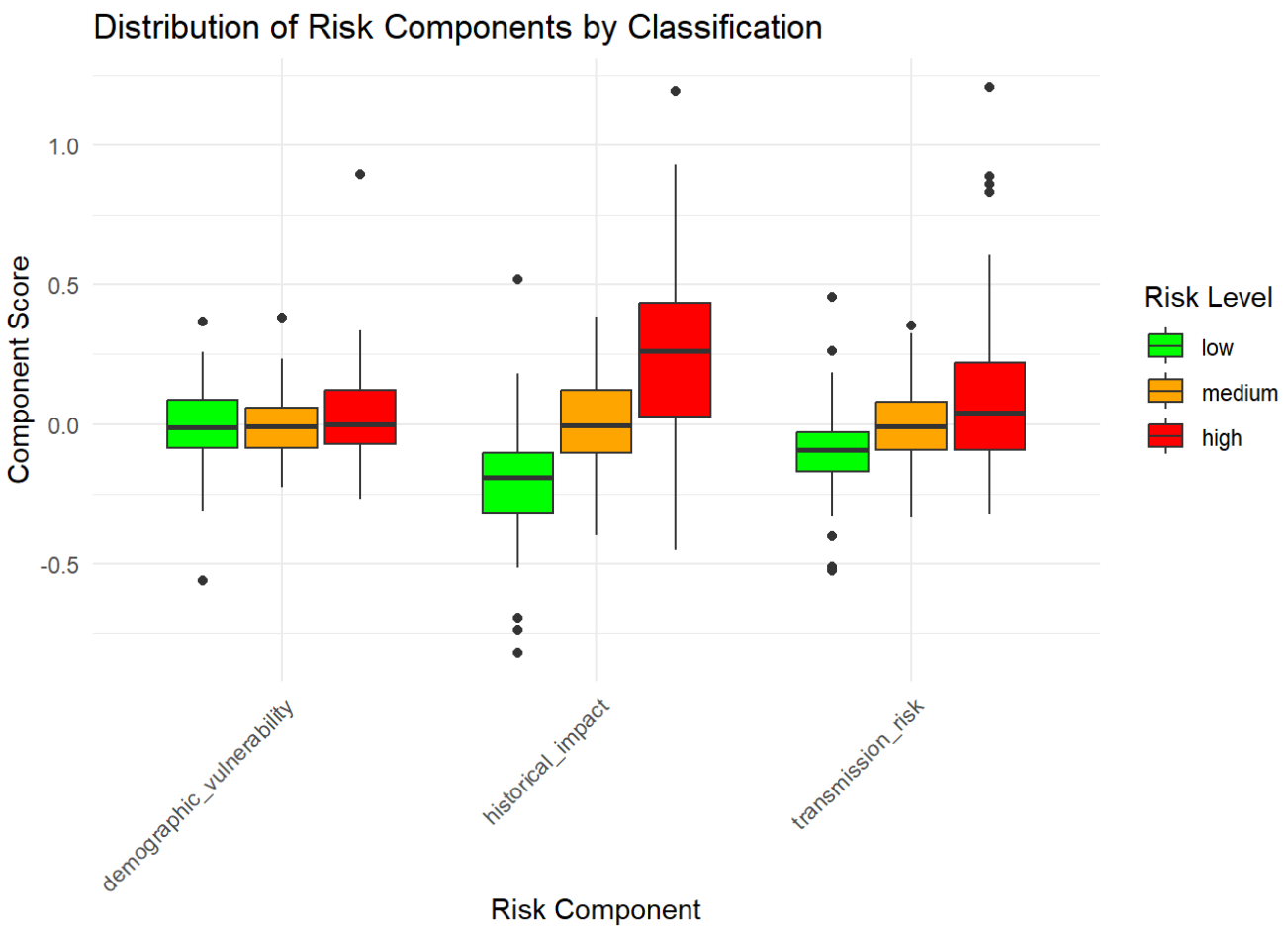

Table 14: Distribution of Risk Components by Classification

The boxplot visualizes how three key components, demographic vulnerability, historical impact, and transmission risk, contribute to COVID-19 risk classification for Texas counties. These components are plotted across three risk levels, providing insight into how different factors drive the overall assessment. Among them, historical impact shows the most pronounced separation, with low-risk counties having strongly negative scores, medium-risk

counties centered near zero, and high-risk counties exhibiting substantially positive values. This suggests that past COVID-19 outcomes—such as case counts, death rates, and fatality ratios—are the strongest predictors of a county's current risk classification. Transmission risk also shows a clear trend, with scores increasing from low to high-risk counties, though the distributions overlap more than in historical impact, indicating some ambiguity in how current transmission patterns differentiate risk levels. Demographic vulnerability, which may include factors like age distribution and pre-existing health conditions, appears relatively consistent across all categories, showing less influence on classification. Since all components are normalized (z-scores), comparisons are based on relative deviation from average values. The gradual progression across all three factors reinforces the validity of using a multi-component model to assess COVID-19 risk, highlighting how both historical and ongoing conditions shape the vulnerability of different counties and inform targeted public health responses.

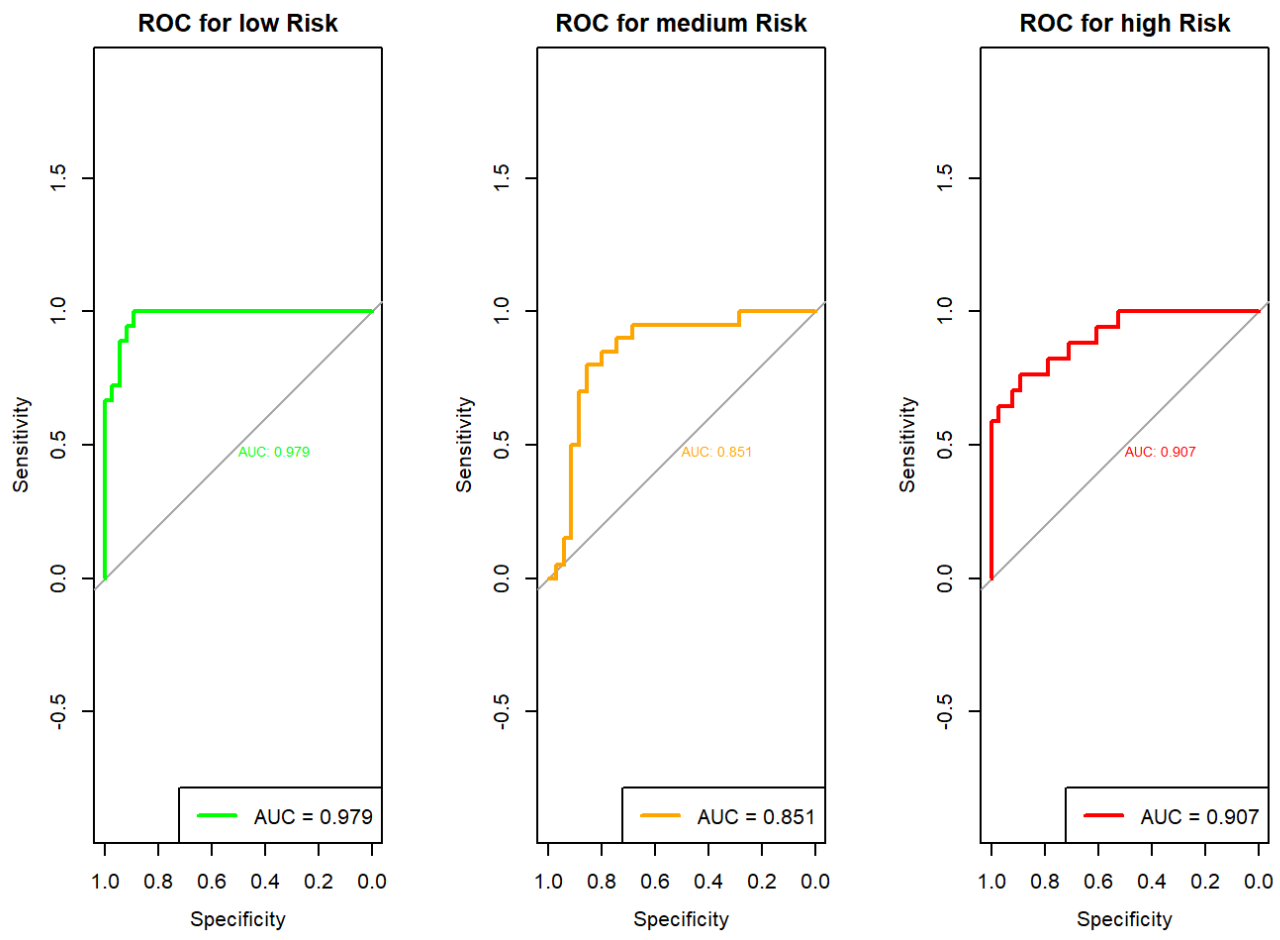

Table 15: ROC for Each Risk Tier

The model's ability to discriminate between counties that truly belong to each risk category is measured using the Area Under the Curve (AUC) for each group. The x-axis describes the false positive rate, and the y-axis is the true positive rate. The grey line is the line of no-discrimination; an AUC of 0.5 indicates random guessing, and the colored curve is the model's actual performance for the respective risk group. A higher AUC indicates that the model more accurately identifies counties at that risk level. For instance, the AUC of 0.979 for the low-risk group suggests the

model is highly effective at distinguishing counties with genuinely low risk from others. Similarly, AUCs of 0.851 for medium risk and 0.907 for high risk indicate good to excellent classification performance in those groups. This means that the model can be reliably used by public health officials or policymakers to classify counties according to their relative health or disease burden risk, which is crucial for targeting interventions, allocating resources, and planning responses tailored to each county's specific needs.

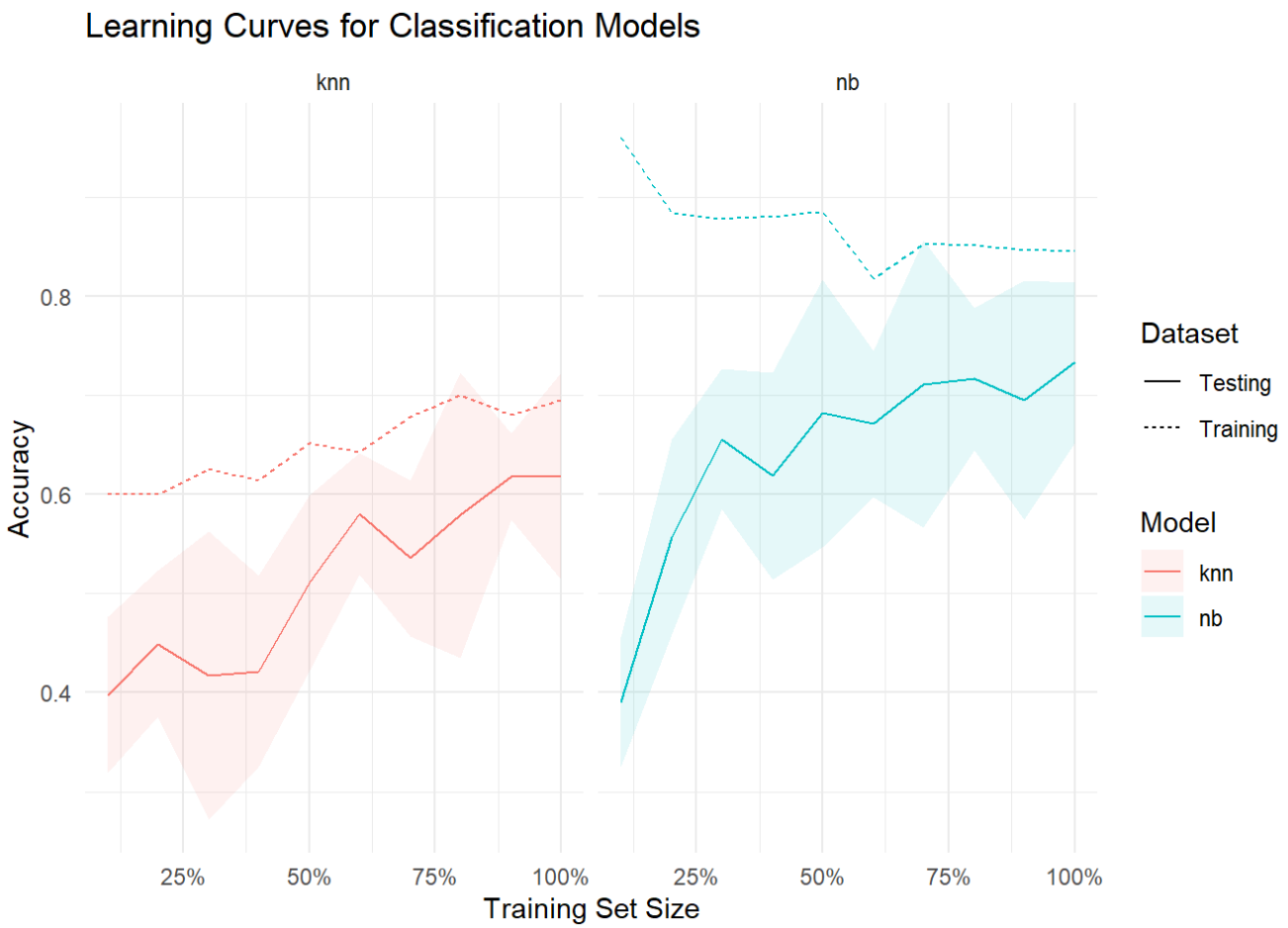

Table 16: Learning Curves for Classification Models

The graph titled "Learning Curves for Classification Models" compares the performance of two machine learning algorithms, k-nearest neighbors and Naive Bayes, in classifying COVID-19 risk levels for Texas counties. The purpose of the learning curves is to illustrate how model accuracy evolves as the size of the training dataset increases, with the x-axis representing training set size (ranging from 25% to 100% of the available data) and the y-axis showing accuracy scores, which span from roughly 40% to 80%. In the visualization, the kNN model is shown in red or pink, while the Naive Bayes model is shown in blue or teal. Each model includes two types of lines: solid lines for testing accuracy, which measures performance on unseen data, and dotted lines for training accuracy, which reflects how well the model fits the training data. Shaded areas around the lines indicate the standard deviation of accuracy across cross-validation folds, providing a measure of variability. The key findings from this graph indicate that Naive Bayes consistently outperforms kNN, achieving higher testing accuracy in the range of 70–75%, while kNN lags behind at

around 60% with greater variance. Additionally, the NB model displays a steeper initial learning curve, implying that it requires less training data to perform well. Furthermore, the smaller gap between its training and testing accuracies suggests that NB is less prone to overfitting compared to kNN. Overall, the visualization highlights that Naive Bayes offers more reliable and stable performance, making it potentially better suited for predicting COVID-19 risk levels in counties.

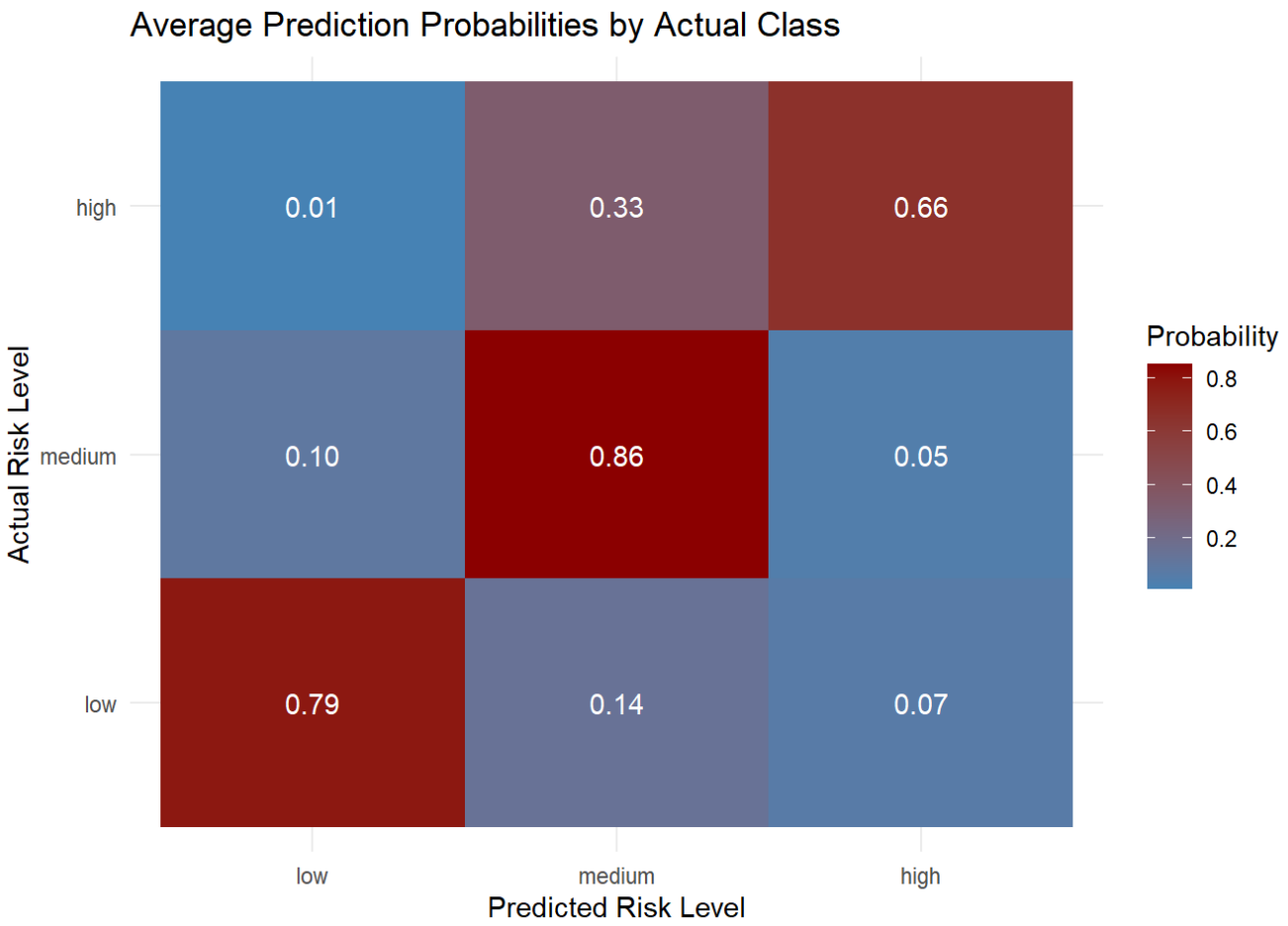

Table 17: Overall Prediction Probabilities from Classification

The illustrates the performance of a COVID-19 risk prediction model by comparing predicted probabilities to the actual risk levels of counties. Along the vertical axis are the actual risk categories. Each cell in the heatmap represents the average probability that the models assign to each predicted category for a given actual risk level, with the rows summing to approximately 1.0. For instance, when a county is actually low risk, the model assigns an average probability of 0.79 to the low-risk category, 0.14 to medium, and 0.07 to high. For medium-risk counties, the model assigns probabilities of 0.10, 0.86, and 0.05 to low, medium, and high, respectively. High-risk counties receive predicted probabilities of 0.01 for low, 0.33 for medium, and 0.66 for high risk. The diagonal values indicate the model's average confidence when its predictions match the actual class. These values suggest that the model is most

accurate in identifying medium-risk counties, followed by low-risk, and is relatively less confident when predicting high-risk counties.

4 Evaluation

The COVID-19 risk assessment model presented here, supported by visualizations and R code, demonstrates strong potential value for policymakers across Texas. The model integrates multiple components—demographic vulnerability, historical impact, and transmission risk, into a comprehensive risk score that classifies counties into low, medium, or high risk. This multifactor approach provides a nuanced understanding of risk, allowing stakeholders to make data-driven decisions tailored to local conditions.

The ROC curves indicate high classification accuracy, especially in low- and high-risk counties, with AUC values of 0.979 and 0.907, respectively. These results show that the model is especially effective at identifying counties that are clearly at low or high risk, which is critical for allocating resources efficiently, such as vaccines, testing kits, or public health messaging. The boxplots of component distributions show clear trends, particularly for historical impact, affirming the model's ability to reflect real-world differences in county-level COVID experiences. The distribution of total risk scores further demonstrates that counties are being grouped meaningfully, even though some overlap near the classification thresholds suggests the need for cautious interpretation in borderline cases.

If used operationally, the model’s value could be assessed by tracking how well it predicts future surges or outcomes, such as increased hospitalizations or mortality rates in high-risk areas. Regular back-testing with updated COVID data could measure its predictive performance over time. Additionally, feedback from on-the-ground stakeholders could help determine whether counties classified as high risk indeed experienced more strain or required more intervention. Ultimately, the model offers a structured, adaptable framework that can help stakeholders prioritize interventions, monitor changing conditions, and respond proactively to emerging threats, making it a valuable decision-support tool in ongoing public health efforts.

5 Deployment

This county-level COVID-19 risk classification model would serve as a decision support tool for public health departments and emergency management agencies. Since this report is targeted for Texas state officials, here is how they can allocate resources.

For regular use cases, health departments in these high-risk counties can be prioritized for receiving medical supplies, testing sites, and vaccine distribution. Mobile vaccination clinics could be strategically deployed to vulnerable communities in medium and high-risk areas. These counties identified as high-risk could implement more stringent public health measures. Medium risk counties could receive targeted educational campaigns about preemptive

behaviors. Low risk counties could maintain baseline monitoring while focusing on sustaining protective factors. The hospitals and other healthcare facilities in high-risk counties would receive early warnings to prepare for possible surge capacity. Staff deployment could be optimized regionally based on predicted risk levels. For public communications, risk level maps would inform residents about local conditions, explaining how to properly combat the virus within their communities and neighborhoods. County officials could use risk level designations to justify and communicate policy decisions.

For a model update schedule and maintenance strategy, to ensure continued accuracy and relevance, the COVID-19 county-level risk classification model will follow a multi-tiered update schedule that incorporates regular data refreshes, periodic model adjustments, and event-driven recalibrations. This structured approach enables the model to remain responsive to dynamic pandemic conditions while preserving scientific rigor and practical usability.

Every two weeks, the model will ingest the most recent data on case counts, mortality rates, vaccination coverage, and population mobility. These updates will generate revised risk classifications, which will be promptly distributed to stakeholders such as public health departments and emergency operations centers. This ensures that decision-makers are working with the most current information available, supporting timely interventions at the county level. On a monthly basis, the model will undergo retraining to account for changes in feature relevance and shifting patterns in pandemic behavior. This includes re-evaluating the importance of input variables—such as testing rates or hospital capacity—and tuning model parameters to optimize performance. Monthly retraining ensures the model adapts to evolving epidemiological trends and maintains its predictive accuracy over time. Once per quarter, the model will be comprehensively evaluated against actual outcomes. This deeper analysis may lead to structural revisions, such as incorporating new data sources (e.g., wastewater surveillance, variant tracking) or adjusting risk thresholds to reflect changes in virus transmissibility, severity, or public health guidance. These quarterly reviews allow the model to stay aligned with the latest scientific findings and public health priorities. The update cycle also includes contingency plans for rapid revision in response to major external events. For instance, the emergence of a new COVID-19 variant would trigger an immediate model review to determine if current variables and thresholds remain valid. Similarly, major policy shifts—such as the lifting of mask mandates or changes in testing protocols—would prompt swift recalibration. Seasonal transitions, which can affect virus spread patterns and health system strain, would also warrant parameter adjustments to maintain accuracy.

Model performance will be continuously monitored using metrics such as accuracy, precision, and recall across each risk category (low, medium, high). These metrics will be compared against real-world outcomes at the county level to assess the model's reliability. Persistent discrepancies or patterns of misclassification will be flagged for further investigation, ensuring systematic errors or biases are quickly identified and addressed.

To maintain methodological soundness and public trust, model outputs will be reviewed by a multidisciplinary panel of experts before each release. Epidemiologists will assess the outputs from a disease dynamics perspective, public health officials will evaluate their practical relevance, and statistical experts will review the model structure and

assumptions. This collaborative oversight ensures that the model remains scientifically credible and operationally useful.

For maximum utility, the model will be fully integrated with existing digital infrastructure. Risk classifications and supporting data will feed directly into state and county health department dashboards, emergency operations centers' decision support systems, and hospital resource management platforms. Additionally, summaries and visualizations will be made available on public-facing COVID-19 information websites to support transparency and public awareness. This integration ensures the model supports both strategic planning and on-the-ground response.

Despite its strengths, users must understand the model's limitations. Predictions inherently carry a degree of uncertainty, particularly in counties with sparse or delayed data reporting. The model may also omit local contextual factors such as policy enforcement, public compliance, and population behavior, which can influence transmission but are difficult to quantify. Therefore, model output should be viewed as one input among several in the decision-making process, not as the sole determinant of public health action. Complementary tools such as community surveillance, direct field observations, and local expertise should be used alongside the model to ensure a comprehensive and context-sensitive response.

6 List of References

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7 Appendix

7.1 Student Contributions

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